

A SIMPLE BATTERY CHARGER With Overcharge Protection

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This simple battery charger, with overcharge protection, can be used to charge multiple cells in a battery. It indicates when the cells are fully charged. The charger requires minimal components, which can be procured easily. The author's circuit wired on a breadboard is shown in Fig. 1.

The circuit of the Li-ion battery charger, shown in Fig. 2, has three parts—charging and balancing

circuit, current limiter, and constant voltage source. The circuit is capable of providing a maximum of 1.5A current and can take input up to 27 volts.

The most important part of the charger is the charging and balancing circuit that incorporates Zener diode TL431, which acts as an adjustable shunt regulator to control the PNP transistor BD140. The TL431 works as a linear regula-

tor whose threshold voltage can be adjusted through trim pot VR1.

Charging and balancing circuit

When the voltage across Zener diode TL431 is below the threshold voltage, the Zener is in off state. Since the base of the transistor is connected to the cathode of TL431, the transistor remains in off stage as well. Therefore, the current flows through the battery, which is connected in parallel, and thus starts charging it.

On the completion of charging, when the voltage of the battery rises over the upper threshold voltage, the TL431 gets activated and connects the base of the transistor to the ground, thus turning the transistor to its conducting state. In this state, the transistor creates a new path for the current to flow, bypassing the battery, and so the charging stops.

The transistor is connected in

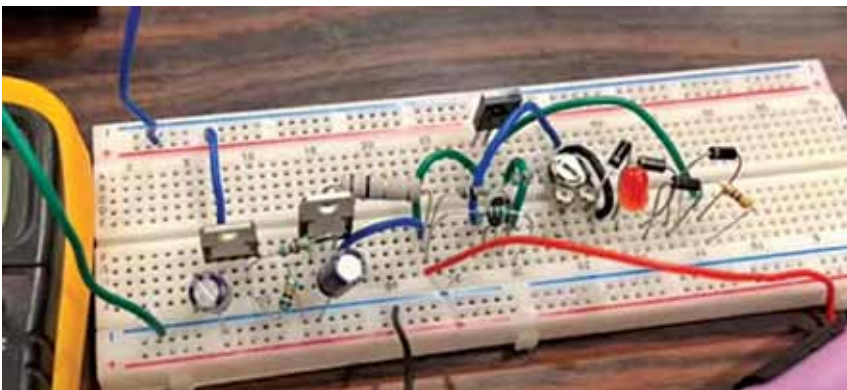


Fig. 1: Author's circuit on breadboard

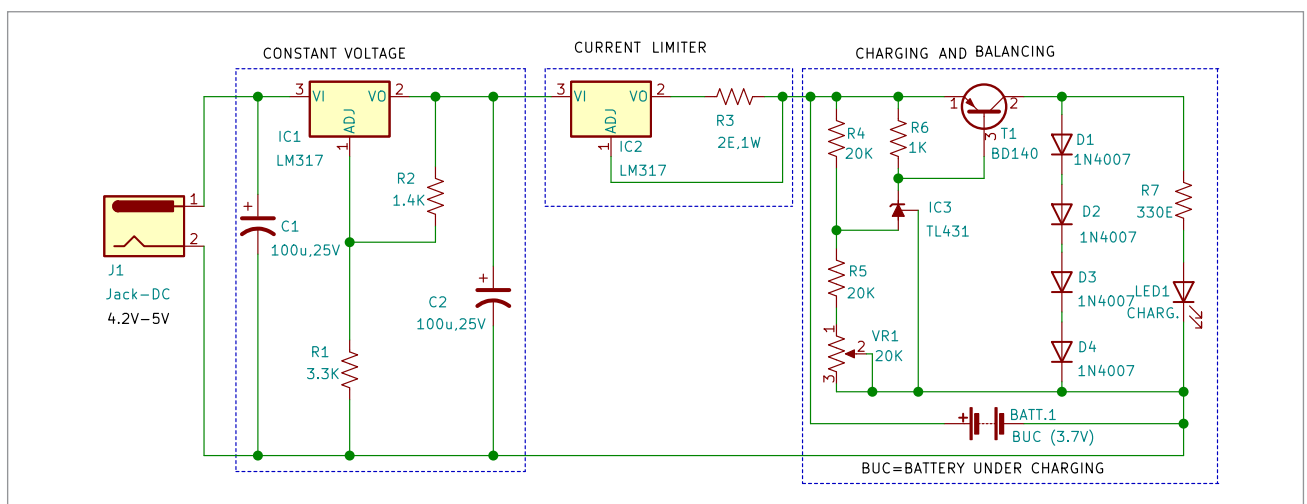


Fig. 2: Circuit diagram of the battery charger